

Cover Letter

New submission to Journal of Cosmology and Astroparticle Physics (JCAP)
Title: Testing Λ CDM with ANN-Reconstructed Expansion History from Cosmic Chronometers
arXiv: 2604.22372 [astro-ph.CO]
Authors: Yuki Hashimoto, Kazuharu Bamba and Sanjay Mandal

Dear Editor(s) (Journal of Cosmology and Astroparticle Physics (JCAP)):

We would like to submit our manuscript entitled “Testing Λ CDM with ANN-Reconstructed Expansion History from Cosmic Chronometers” (arXiv: 2604.22372 [astro-ph.CO]) to Journal of Cosmology and Astroparticle Physics (JCAP).

In this paper, we propose a novel artificial neural network (ANN)-based framework for the non-parametric reconstruction of the late-time cosmic expansion. The framework is trained and validated through a three-stage screening pipeline prior to its application to real observational data. As a demonstration of its effectiveness, we reconstruct the Hubble parameter using the latest cosmic chronometer measurements. Our results show that the reconstructed expansion history aligns with the predictions of the Λ CDM model within observational uncertainties, thereby supporting the robustness and reliability of the proposed approach.

We would like to sincerely appreciate your kind consideration.

Yours sincerely,

Yuki Hashimoto, Kazuharu Bamba and Sanjay Mandal